

SOFTWARE TESTING ASSESSMENT REPORT

Project: Agro India — Digital Agricultural Platform & E-Commerce

Target Platform: Agro India Web & Mobile App

Evaluation Focus: Test Strategy, CRUD & Risk Analysis

Domain: Agri-Tech & Input

Release Status: Pre-Production Release

PART 1: TEST CASE DESIGN

1. USER REGISTRATION & FARMER PROFILES

TC_REG_01 — Farmer Mobile / Email Onboarding (Positive)

Register a new farmer account using a valid 10-digit mobile number or email, verifying OTP/password authentication and successful profile record creation.

TC_REG_02 — Malformed Contact Information (Negative)

Attempt registration with a malformed phone number (fewer than 10 digits) or invalid email structure and verify inline validation blocks submission.

TC_REG_03 — Regional Multilingual Language Preference (Edge)

Register while selecting a regional Indian language preference (e.g., Hindi, Telugu, Punjabi) and verify that the UI updates instantly upon activation.

2. USER LOGIN & ROLE-BASED ACCESS

TC_LOG_01 — Role-Based Dashboard Redirection (Positive)

Log in with verified credentials and ensure redirection to the appropriate dashboard based on assigned user role (Farmer, Distributor, or Admin).

TC_LOG_02 — Unverified Credentials Handling (Negative)

Attempt login with incorrect passwords or unverified phone numbers and verify that authentication fails safely with generic alerts.

TC_LOG_03 — Inactive Session Timeout Protection (Edge)

Verify that inactive user sessions expire securely after a set threshold to protect sensitive account and transaction profiles.

3. AGRO-INPUT / PRODUCT CRUD OPERATIONS

TC_CRUD_01 — Agro-Input Catalog Creation (Positive)

Add a new agricultural product (Bio-pesticides, Fertilizers, Seeds) with valid name, category, pricing, stock quantity, and imagery; verify catalog indexing.

TC_CRUD_02 — Negative Pricing and Missing SKU Validation (Negative)

Attempt to create a product listing with a negative price or missing mandatory SKU fields and verify the database rejects the entry.

TC_CRUD_03 — Catalog Search & Category Filtration (Positive)

Browse and filter product catalogs by categories (Plant Growth Regulators, Organic Carbon) or search keywords to ensure accurate query retrieval.

TC_CRUD_04 — Inventory Stock & Price Adjustment (Update)

Modify stock levels or adjust pricing for an agro-product and verify updates reflect instantly for front-end buyers.

TC_CRUD_05 — Discontinued Product Removal (Delete)

Remove a discontinued product or crop advisory record and ensure it is safely soft-deleted or removed from active public listings.

4. INPUT VALIDATION & AGRICULTURAL SPECIFICS

TC_VAL_01 — Inventory Stock Bounds Verification (Negative)

Order an amount of fertilizer exceeding available inventory stock limits and verify that checkout flags a stock-shortage error.

TC_VAL_02 — Regional Character Search Injection (Edge)

Input regional characters or special symbols into the product or crop disease search bar to ensure robust text sanitization and no server crashes.

PART 2: BUG IDENTIFICATION & RISK AREAS

Potential high-risk failure modes identified for pre-production evaluation without executing the application:

BUG/RISK 01 — Inaccurate Floating-Point Calculations

Severity: Major

Reason & Operational Impact: Agricultural items are sold in bulk quantities and fractional weights. Rounding errors can corrupt invoice totals for large farmer orders.

BUG/RISK 02 — Broken Object Level Authorization (BOLA)

Severity: Critical

Reason & Operational Impact: Weak authorization checks could allow registered farmers to view or alter another farmer's order history or invoices by modifying URL IDs.

BUG/RISK 03 — Multilingual Text Overflows

Severity: Minor

Reason & Operational Impact: Rendering Indian regional text can trigger layout wrap issues or overlapping buttons on mobile browsers if Unicode fonts aren't bounded.

BUG/RISK 04 — Inventory Race Conditions

Severity: Critical

Reason & Operational Impact: Concurrent purchase attempts for limited seasonal stock can permit duplicate checkouts, causing negative inventory states and fulfillment failure.

BUG/RISK 05 — Unthrottled OTP Gateway Requests

Severity: Major

Reason & Operational Impact: Lack of rate limiting on SMS requests leaves endpoints vulnerable to spam scripts, depleting API credits and locking out genuine users.

BUG/RISK 06 — Unsanitized Crop Disease Uploads

Severity: Critical

Reason & Operational Impact: Permitting file uploads without rigorous extension checks enables attackers to execute malicious scripts disguised as image formats.

BUG/RISK 07 — Timezone Desynchronization

Severity: Minor

Reason & Operational Impact: Server time mismatches can cause flash sales, weather alerts, or time-sensitive harvest advisory schedules to display erroneous local hours.

BUG/RISK 08 — SQL Injection in Search Filters

Severity: Critical

Reason & Operational Impact: Inadequately parameterized query strings used for searching fertilizers or inputs can expose the underlying database to total compromise.